PAPER ID-310776

Subject Code: KME072

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VII) THEORY EXAMINATION 2023-24
HVAC SYSTEMS

TIME: 3 HRS **M.MARKS: 100**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.
2. Use of Psychrometric chart is permitted.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Qno.	Question	Marks	CO
a.	Define subcritical and transcritical cycles.	2	1
b.	What is the greenhouse effect? Name the gases causing this effect and their sources.	2	1
c.	Explain the Grand Sensible Heat Factor.	2	2
d.	Discuss effective temperature. Name any two factors affecting it.	2	2
e.	Classify airconditioning systems	2	3
f.	What is a heat pump? Write an expression to calculate the performance index of a heat pump.	2	3
g.	Define evaporative cooling. Draw this process on a skeleton psychrometric chart.	2	4
h.	What is Sol-Air temperature? List the factors affecting it.	2	4
i.	Define Throw and Drop in air distribution.	2	5
j.	Name the key elements of an HVAC control system.	2	5

SECTION B

2. Attempt any *three* of the following: **10 x 3 = 30**

a.	Discuss the effect of the increase in evaporator and condenser temperatures on the performance of the vapour compression refrigeration system.	10	1
b.	30 m^3/min of a stream of moist air at 15ºC DBT and 13ºC WBT is mixed with 12 m^3/min of a second stream at 25ºC DBT and 18ºC WBT. Barometric pressure is one atmosphere. Calculate the dry bulb temperature and wet bulb temperature of the resulting mixture.	10	2
c.	Differentiate between unitary air-conditioning and central air-conditioning.	10	3
d.	Explain the infiltration of air and its causes.	10	4
e.	Explain the following (a) Classification of ducts (b) Types of centrifugal fans	10	5

SECTION C

3. Attempt any *one* part of the following: **10 x 1 = 10**

a.	Write short notes on: (i) Ejector Refrigeration Cycle (ii) VCR cycle with subcooling and superheating	10	1
b.	Explain the desirable properties of an ideal refrigerant. What are future refrigerants?	10	1

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4.	**Attempt any *one* part of the following:**	**10 x 1 = 10**	
a.	Atmospheric air at a dry bulb temperature of 16°C and 25% relative humidity passes through a furnace and then through a humidifier, in such a way that the final dry bulb temperature is 30°C and 50% relative humidity. Determine the heat and moisture added to the air. Also, calculate the sensible heat factor of the process.	10	2
b.	What is human comfort? Discuss the factors affecting human comfort.	10	2

5.	**Attempt any *one* part of the following:**	**10 x 1 = 10**	
a.	Describe ventilation and its purpose. Explain how a natural ventilation system is different from mechanical ventilation.	10	3
b.	Discuss the all-water, air-water, and all-water type airconditioning systems.	10	3

6.	**Attempt any *one* part of the following:**	**10 x 1 = 10**	
a.	A conference room for seating 100 persons is to be maintained at 22 ^{0}C D.B.T. and 60% R.H. The outdoor conditions are 40 °C D.B.T. and 27 ^{0}C W.B.T. The various loads in the conference room are as follows: Sensible latent heat load per person: 80 W Latent heat load per person: 50 W Lights and fans: 15000 W Sensible heat gain through glass, walls, ceiling, etc.: 15000 W. Rate of air infiltration: 20 m^3/min. Determine the room sensible heat factor.	10	4
b.	Discuss the procedure to estimate the cooling load using a suitable example.	10	4

7.	**Attempt any *one* part of the following:**	**10 x 1 = 10**	
a.	The main air supply duct of an air-conditioning system is 800 mm x 600 mm in cross-section and carries 300 m^3/min of standard air. It branches into two ducts of cross-section 600 mm x 500 mm and 600 mm x 400 mm. If the mean velocity in the larger branch is 480 m/min, calculate; (i) Mean velocity in the main duct and (ii) Mean velocity pressure in each duct.	10	5
b.	Compare the characteristics of backward-curved and forward-curved blade fans with the help of suitable sketches.	10	5

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